



BAZAARO HYPERLOCAL MARKETPLACE TESTING REPORT

SE3040 – Application Frameworks

1. Introduction

This report presents the testing activities performed for the **Hyperlocal Marketplace for Products & Services** system. The system allows users to browse nearby products and services, create listings, place orders, manage bookings, and interact through a role-based platform

2. Testing Objectives

The objective of testing was to:

- Verify core system functionality
- Validate API behavior
- Ensure correct access control and validation
- Test key user workflows such as orders and bookings
- Perform a basic performance test

3. Scope of Testing

The following modules were tested:

- Authentication
- Categories
- Product Listings
- Product Orders
- Service Listings
- Service Bookings
- Admin/Moderation
- UI Routes and Protected Access
- Performance Testing

Out of scope:

- Full UI end-to-end testing
- Payment provider testing
- Security and penetration testing
- Cross-browser testing
- Advanced performance benchmarking.

4. Testing Types Used

Type	Description
Functional Testing	Validation of system behavior
API Testing	Testing backend endpoints
UI Testing	Basic route validation
Performance Testing	Load testing using k6

5. Testing Tools Used

- **Playwright** → UI + API + integration testing
- **k6** → Performance testing
- **Swagger/OpenAPI** → API documentation

6. Testing Environment

- OS: Windows 10 (64-bit)
- Backend: `http://localhost:5000/api`
- Frontend: `http://localhost:3000`
- Playwright workers: 1
- Test directory: `tests/`

7.Test Cases Summary

Total test cases: **27**

- Playwright tests: 26
- Performance tests: 1

Module	Tests
Auth	2
Categories	3
Product Listings	4
Product Orders	3
Service Listings	4
Service Bookings	3
Admin	3
UI Routes	4
Performance	1

8.Test Cases Execution Table

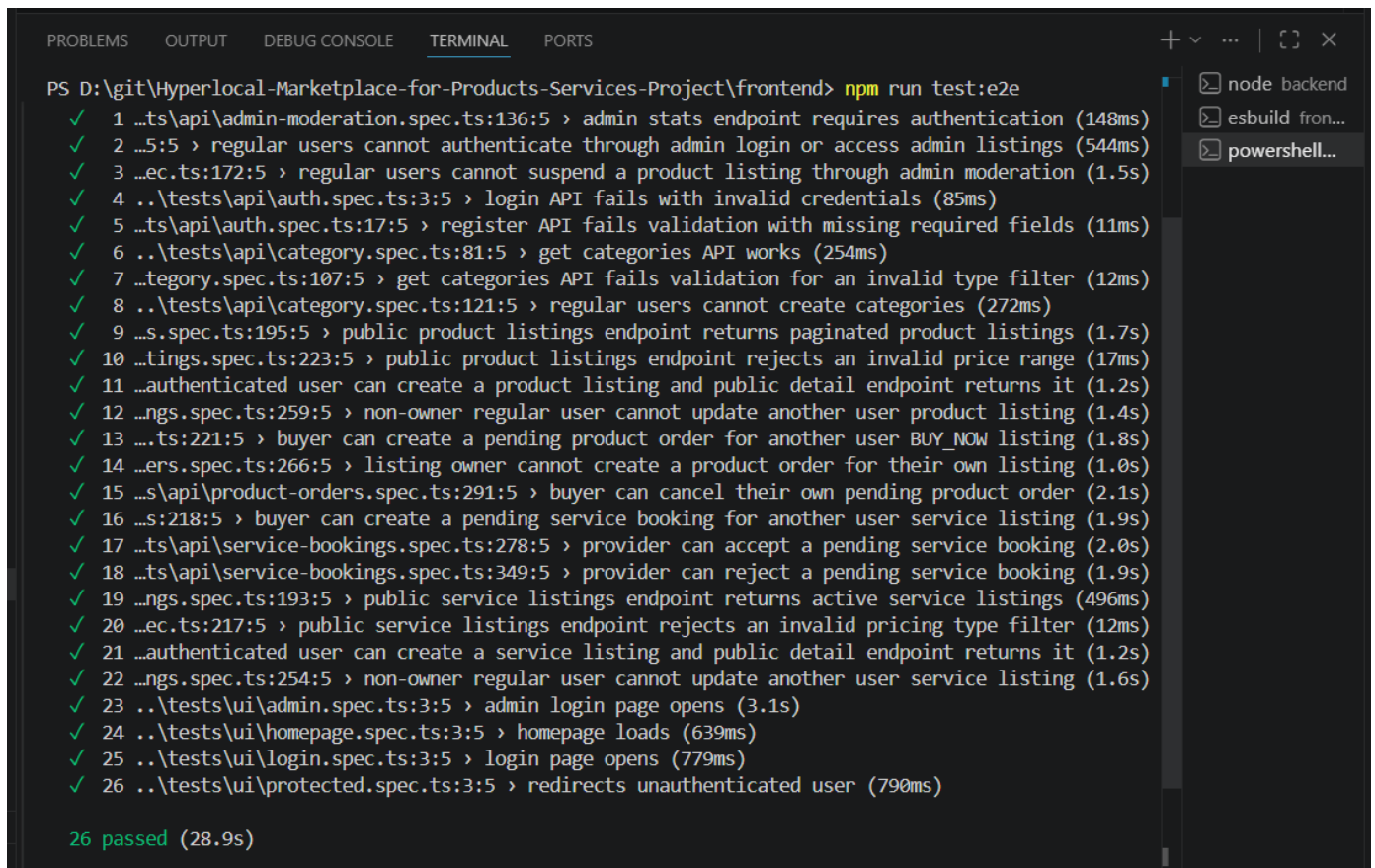
Test Case ID	Scenario	Expected Result	Actual Result	Status
TC-01	Homepage loads	Page loads successfully	Page loaded correctly	Pass
TC-02	Login page opens	Login page visible	Login page displayed	Pass
TC-03	Invalid login	Error message shown	Correct error shown	Pass
TC-04	Get categories API	Returns 200 + data	Returned valid data	Pass
TC-05	Create product listing	Listing created	Listing successfully created	Pass
TC-06	Place order	Order created	Order created successfully	Pass
TC-07	Create booking	Booking created	Booking created	Pass
TC-08	Accept booking	Status updated	Status changed correctly	Pass
TC-09	Admin access control	Unauthorized blocked	Access denied correctly	Pass

Note: The above table shows a sample of executed test cases. The complete list of test cases is documented in the Test Cases section.

9. Test Execution

1. Playwright Execution

- ❖ Command Used: **npm run test:e2e**
- ❖ Result: **All tests passed successfully.**



```
PS D:\git\Hyperlocal-Marketplace-for-Products-Services-Project\frontend> npm run test:e2e
✓ 1 ...ts\api\admin-moderation.spec.ts:136:5 › admin stats endpoint requires authentication (148ms)
✓ 2 ...5:5 › regular users cannot authenticate through admin login or access admin listings (544ms)
✓ 3 ...ec.ts:172:5 › regular users cannot suspend a product listing through admin moderation (1.5s)
✓ 4 ..\tests\api\auth.spec.ts:3:5 › login API fails with invalid credentials (85ms)
✓ 5 ...ts\api\auth.spec.ts:17:5 › register API fails validation with missing required fields (11ms)
✓ 6 ..\tests\api\category.spec.ts:81:5 › get categories API works (254ms)
✓ 7 ...tegory.spec.ts:107:5 › get categories API fails validation for an invalid type filter (12ms)
✓ 8 ..\tests\api\category.spec.ts:121:5 › regular users cannot create categories (272ms)
✓ 9 ...s.spec.ts:195:5 › public product listings endpoint returns paginated product listings (1.7s)
✓ 10 ...tings.spec.ts:223:5 › public product listings endpoint rejects an invalid price range (17ms)
✓ 11 ...authenticated user can create a product listing and public detail endpoint returns it (1.2s)
✓ 12 ...ngs.spec.ts:259:5 › non-owner regular user cannot update another user product listing (1.4s)
✓ 13 ...ts:221:5 › buyer can create a pending product order for another user BUY_NOW listing (1.8s)
✓ 14 ...ers.spec.ts:266:5 › listing owner cannot create a product order for their own listing (1.0s)
✓ 15 ...s\api\product-orders.spec.ts:291:5 › buyer can cancel their own pending product order (2.1s)
✓ 16 ...s:218:5 › buyer can create a pending service booking for another user service listing (1.9s)
✓ 17 ...ts\api\service-bookings.spec.ts:278:5 › provider can accept a pending service booking (2.0s)
✓ 18 ...ts\api\service-bookings.spec.ts:349:5 › provider can reject a pending service booking (1.9s)
✓ 19 ...ngs.spec.ts:193:5 › public service listings endpoint returns active service listings (496ms)
✓ 20 ...ec.ts:217:5 › public service listings endpoint rejects an invalid pricing type filter (12ms)
✓ 21 ...authenticated user can create a service listing and public detail endpoint returns it (1.2s)
✓ 22 ...ngs.spec.ts:254:5 › non-owner regular user cannot update another user service listing (1.6s)
✓ 23 ..\tests\ui\admin.spec.ts:3:5 › admin login page opens (3.1s)
✓ 24 ..\tests\ui\homepage.spec.ts:3:5 › homepage loads (639ms)
✓ 25 ..\tests\ui\login.spec.ts:3:5 › login page opens (779ms)
✓ 26 ..\tests\ui\protected.spec.ts:3:5 › redirects unauthenticated user (790ms)

26 passed (28.9s)
```

2. .Playwright HTML Report

❖ Generated Using: **npm run test:e2e:report**

Q

Search tests

All26

✓ Passed26

Failed0

Flaky0

Skipped0

4/11/2026, 12:01:37 AMTotal time: 28.9s

▼api/admin-moderation.spec.ts

✓admin stats endpoint requires authentication148ms

api/admin-moderation.spec.ts:136

✓regular users cannot authenticate through admin login or access admin listings544ms

api/admin-moderation.spec.ts:145

✓regular users cannot suspend a product listing through admin moderation1.5s

api/admin-moderation.spec.ts:172

▼api/auth.spec.ts

✓login API fails with invalid credentials85ms

api/auth.spec.ts:3

✓register API fails validation with missing required fields11ms

api/auth.spec.ts:17

▼api/category.spec.ts

✓get categories API works254ms

api/category.spec.ts:81

✓get categories API fails validation for an invalid type filter12ms

api/category.spec.ts:107

✓regular users cannot create categories272ms

api/category.spec.ts:121

▼api/product-listings.spec.ts

✓public product listings endpoint returns paginated product listings1.7s

api/product-listings.spec.ts:195

✓public product listings endpoint rejects an invalid price range17ms

api/product-listings.spec.ts:223

✓authenticated user can create a product listing and public detail endpoint returns it1.2s

api/product-listings.spec.ts:238

✓non-owner regular user cannot update another user product listing1.4s

api/product-listings.spec.ts:259

▼api/product-orders.spec.ts

✓buyer can create a pending product order for another user BUY_NOW listing1.8s

api/product-orders.spec.ts:221

✓listing owner cannot create a product order for their own listing1.0s

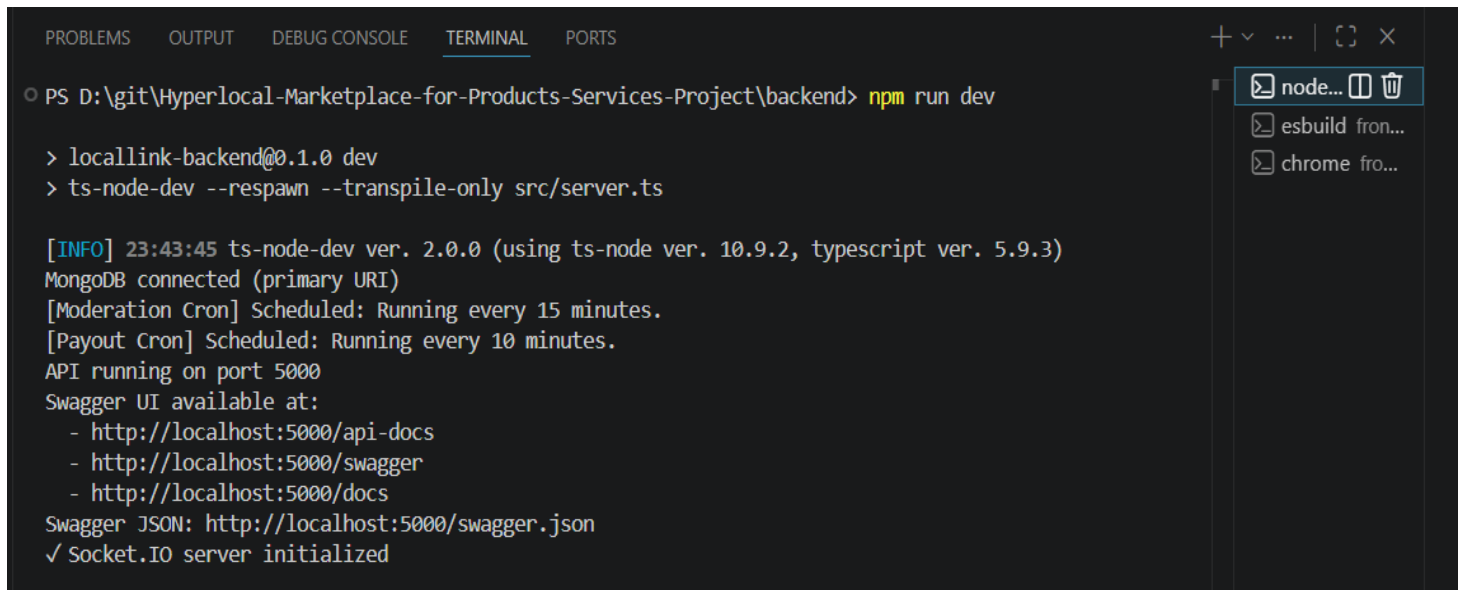
api/product-orders.spec.ts:266

✓buyer can cancel their own pending product order2.1s

api/product-orders.spec.ts:291

✓ api/service-bookings.spec.ts		
✓	buyer can create a pending service booking for another user service listing api/service-bookings.spec.ts:218	1.9s
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✓	non-owner regular user cannot update another user service listing api/service-listings.spec.ts:254	1.6s
✓ ui/admin.spec.ts		
✓	admin login page opens ui/admin.spec.ts:3	3.1s
✓ ui/homepage.spec.ts		
✓	homepage loads ui/homepage.spec.ts:3	639ms
✓ ui/login.spec.ts		
✓	login page opens ui/login.spec.ts:3	779ms
✓ ui/protected.spec.ts		
✓	redirects unauthenticated user ui/protected.spec.ts:3	790ms

3. Backend Running



```
PS D:\git\Hyperlocal-Marketplace-for-Products-Services-Project\backend> npm run dev

> locallink-backend@0.1.0 dev
> ts-node-dev --respawn --transpile-only src/server.ts

[INFO] 23:43:45 ts-node-dev ver. 2.0.0 (using ts-node ver. 10.9.2, typescript ver. 5.9.3)
MongoDB connected (primary URI)
[Moderation Cron] Scheduled: Running every 15 minutes.
[Payout Cron] Scheduled: Running every 10 minutes.
API running on port 5000
Swagger UI available at:
- http://localhost:5000/api-docs
- http://localhost:5000/swagger
- http://localhost:5000/docs
Swagger JSON: http://localhost:5000/swagger.json
✓ Socket.IO server initialized
```

10. Performance Testing (k6)

❖ Command Used: **k6 run tests/performance/basic-load.js**

Test details:

- Endpoint: /api/categories
- Virtual Users: 10
- Duration: 10 seconds
- Requests: 83
- Failures: 0
- Avg Response Time: ~252 ms

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS D:\git\Hyperlocal-Marketplace-for-Products-Services-Project> C:\k6\k6.exe run tests/performance/basic-load.js

✓ status is 200

HTTP
http_req_duration.....: avg=252.83ms min=104.69ms med=218.23ms max=928.6ms p(90)=392.22m
s p(95)=576.29ms
{ expected_response:true }...: avg=252.83ms min=104.69ms med=218.23ms max=928.6ms p(90)=392.22m
s p(95)=576.29ms
http_req_failed.....: 0.00% 0 out of 83
http_req_failed.....: 0.00% 0 out of 83
http_reqs.....: 83 7.416802/s

EXECUTION
iteration_duration.....: avg=1.25s min=1.1s med=1.21s max=1.93s p(90)=1.4s
p(95)=1.58s
iterations.....: 83 7.416802/s
vus.....: 2 min=2 max=10
vus_max.....: 10 min=10 max=10

NETWORK
data_received.....: 6.0 MB 534 kB/s
data_sent.....: 7.0 kB 623 B/s

running (11.2s), 00/10 VUs, 83 complete and 0 interrupted iterations
○ default ✓ [=====] 10 VUs 10s
PS D:\git\Hyperlocal-Marketplace-for-Products-Services-Project>
```

11. Bug Summary

- No unresolved critical bugs found
- All issues encountered during setup were resolved
- No blocking defects remain in tested features

12. Conclusion

The system has been successfully tested for core functionalities. All implemented test cases passed, and performance testing showed stable behavior under basic load conditions. The application is considered functionally stable for the tested scenarios.

